NOOR SYED

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PROFILE

Data Scientist with 3+ years of experience in causal inference, A/B experimentation, and end-to-end data pipeline development. Proven ability to identify growth levers from large-scale datasets and translate quantitative findings into actionable strategy for senior stakeholders. Background in Economics, Mathematics & Computer Science, with deep expertise in experimentation design, user segmentation, and scalable data pipelines — oriented toward driving subscriber growth across diverse, high-growth markets.

EXPERIENCE

Research Data Scientist | **LUMS Learning Institute** | Lahore, Pakistan *Aug 2025 – May 2026*

- Designed and executed a university-wide causal study using Propensity Score Matching (ML-based, Python & STATA) to evaluate the impact of attendance on student performance — direct application of experimentation methodology to measure initiative impact.
- Managed end-to-end data pipeline: survey design, data cleaning, feature engineering, and quantitative modelling for a mixed-methods study with 5,000+ student records.
- Built dashboards and visualisations to communicate findings to academic and non-specialist stakeholders, distilling complex causal results into clear, actionable narratives.
- Conducted systematic literature reviews under Dr. Hadia Majid, Dr. Ummad Mazhar, and Dr. Malik Jahan Khan.

Data Scientist | **CERP – EdTech Hub (RCT, 560 Schools)** | Lahore, Pakistan *Jun – Aug 2025*

Collaborated with provincial government partners on a large-scale RCT evaluating a tech-supported foundational learning programme, under Prof. Christina Brown (University of Chicago).

- Engineered longitudinal data pipelines on 80,000+ observations, improving dataset merge rates from 47% to 95% — enabling reliable, scalable performance tracking across a complex, multi-school dataset.
- Applied Double Machine Learning (DRLearner / Random Forests) to estimate Average Treatment Effects, controlling for teacher fixed effects and confounders — identifying causal growth levers from observational and experimental data.
- Conducted heterogeneity analysis by performance quartile, gender, and class level using Bonferroni-corrected regressions to produce a data-driven resource allocation hierarchy — equivalent to subscriber segmentation for growth strategy.
- Implemented K-Means clustering for user segmentation; translated segment profiles into targeted intervention recommendations, mirroring subscriber segmentation for channel optimisation.
- Built interactive analytics dashboards (Power BI / Python) communicating causal findings to government stakeholders, distilling complex results into clear recommendations for non-technical decision-makers.

Teaching Assistant – Data Science | **LUMS** | Lahore, Pakistan *Jan – Jul 2025*

- Designed and delivered Python-based data analysis training to 100+ undergraduates: pandas, scikit-learn, statsmodels, and applied ML.
- Created problem sets focused on real-world data interpretation, statistical modelling, and ML pipelines.

Data Analyst – WASH & Impact Evaluation | **Tayaba Organization (with Reckitt)** | Rajanpur, Pakistan *Jun – Sep 2024*

- Led primary data collection, needs assessments, and impact evaluation for a humanitarian emergency response; built dashboards and a Situational Report informing interventions that reached 3,986 beneficiaries.
- Co-authored a proposal recognised at COP28 and awarded the Impact Hero 2025 Award; synthesised analytical findings into policy-relevant narratives for decision-makers.

Research Assistant | **University of Lahore, Dept. of Economics** | Lahore, Pakistan *Aug – Nov 2023*

- Analysed survey data from 11,650+ respondents (STATA, Python) across two HEC-funded projects; applied linear regression to identify determinants of women's earnings and co-authored a policy brief linked to SDG 5.

SELECTED PROJECTS

Discipline & Human Capital Formation | **Senior Thesis** | *Aug – Dec 2025*

- Econometric + ML study on UNICEF MICS data (15,590 children): ordered logit, one-vs-rest Random Forest with SMOTE; validated feature importance with permutation and LOCO. CGPA 3.9 semester.

Pakistan Demographic Dividend | **Time Series Analysis** | *Aug – Dec 2025*

- VECM and mediation analysis in R on demographic structure's impact on GDP growth; bootstrapped IRFs for robustness, identifying weak capital formation as binding constraint.

Student Performance Analytics | **ML Study** | *Aug – Dec 2024*

- Analysed 50+ performance factors; built Random Forest and Gradient Boosting models; applied KMeans clustering to segment at-risk students; used SHAP for model interpretability.

EDUCATION

BSc (Hons.) Economics, Mathematics & Computer Science | **LUMS** | Lahore, Pakistan *Sep 2022 – May 2026*

- CGPA: 3.83 | Semester CGPA: 3.90 | Dean's Honour List (Top 5%, multiple semesters)
- Relevant coursework: Deep Learning · Foundations of AI & ML · Data Science · Econometrics I & II · Advanced Microeconomics · Linear Algebra · Operations Research · Probability & Statistics · Real Analysis

SKILLS

Languages & Data: Python (pandas, NumPy, scikit-learn, statsmodels, PyTorch, TensorFlow, XGBoost, LightGBM) · R (tidyverse, ggplot2, VECM) · SQL · STATA/SE · C++

Experimentation & Causal Inference: A/B Testing · RCT Design · Difference-in-Differences · Regression Discontinuity · Instrumental Variables · Propensity Score Matching · Double Machine Learning · Panel Data · VECM

Growth & Segmentation: User Segmentation (K-Means / Hierarchical Clustering) · Treatment Effect Estimation · Heterogeneity Analysis · Conversion & Retention Analytics

ML & Deep Learning: Random Forests · Gradient Boosting · Neural Networks (MLP, CNN, RNN, Transformer) · SHAP · LOCO · Cross-Validation

Visualisation & BI: Power BI · Tableau · Matplotlib · Seaborn · ggplot2 · QGIS

Cloud & Platforms: GCP/BigQuery (familiar) · Python production pipelines · Excel Advanced / VBA

AI Tools: GitHub Copilot · Claude · LLM-assisted development workflows

HONOURS & CERTIFICATIONS

- 25 Under 25 Award – Australian High Commission
- Dean's Honour List – LUMS (Top 5%, multiple semesters)
- Google Data Analytics Individual Qualification (Jun 2023)
- Responsible Conduct of Research (RCR) – CITI Program (Jun 2025)
- Human Subjects Research SBE – CITI Program (Aug 2025)